H→WW · 2022POSTEE · V11 6-CLASS MVA

Kinematic cuts buy pre-selection S/√B — they do not buy control-region purity

The MVA defines **purer** control regions *without* any kinematic cut, and already applies those cuts internally when it picks the signal region.
The real lever on SR yield is the **charm-tag working point**.

01 · The $t\bar{t}$ control region is purer when the MVA defines it

Purity = fraction of events whose *true* process is $t\bar{t}$. Pooled MC, 4,046,127 events, raw/unweighted.

$T\bar{T}$ CR DEFINITION	N	$T\bar{T}$ PURITY	SIGNAL CONTAM.	NON-$T\bar{T}$ BKG
argmax = $t\bar{t}$, no kinematic cuts	**1,565,461**	**87.86%**	**0.0004%**	**12.14%**
$mT_{l2}>30$ & $mT_{ll}\leq 60$ *(classic top CR)*	565,368	78.10%	0.0142%	21.89%
$m_{ll}>72$	2,375,781	83.04%	0.0009%	16.96%
$m_{ll}>72$ & $mT_{l2}>30$ & $mT_{ll}>60$	1,614,248	83.12%	0.0004%	16.88%

The MVA CR wins on both axes at once — purest (87.9% vs 78.1%) *and* 2.8× larger (1.57M vs 565k). The classic cut-based top CR is the **worst of the four**, and carries 35× the signal contamination of the MVA CR.

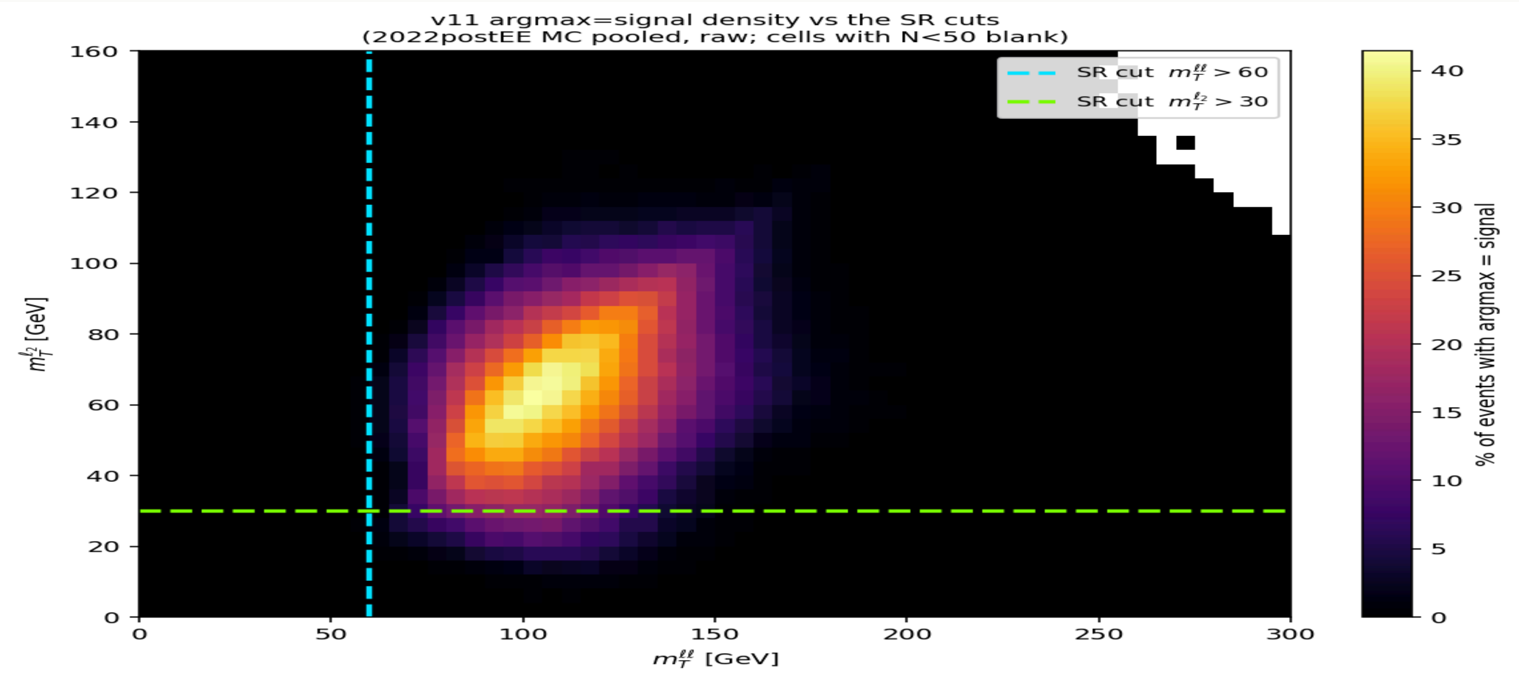
01b · What is actually in the MVA $t\bar{t}$ CR

TRUE PROCESS	N	FRACTION
$t\bar{t}$	1,375,464	87.86%
single top	146,891	9.38%
Higgs bkg	41,122	2.63%
diboson	1,872	0.12%
V+jets	105	0.01%
H+c (signal)	**7**	**0.00%**

> Single-top is the only real contaminant — and it is physically $t\bar{t}$-like, which is exactly what you want a $t\bar{t}$ CR to constrain. Signal leakage is **7 events out of 1.57M**.

02 · The MVA has already internalised the SR kinematic cuts

Fraction of events called signal across the (m_{ll}, mT_{ll}) plane — no kinematic cut applied



BELOW THE MT_{LL} WALL

0.0098%

75 events of 662,103

ABOVE THE WALL

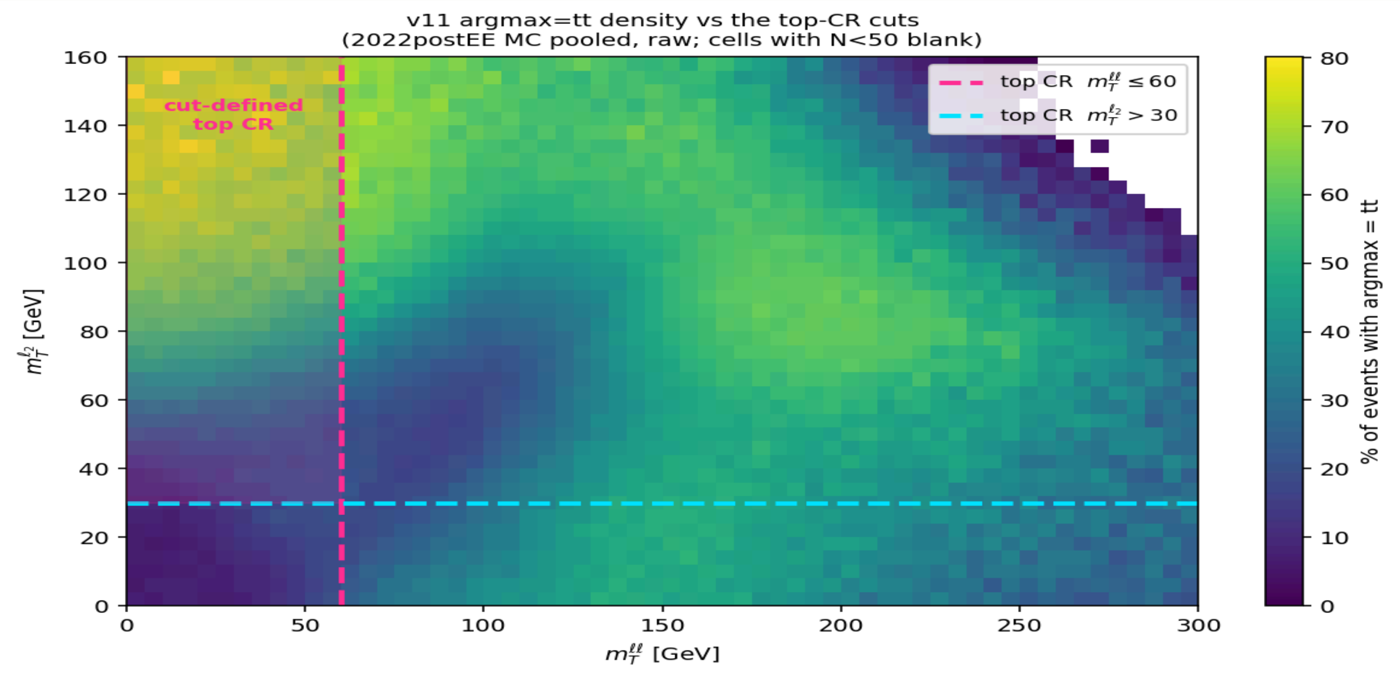
11.36%

1,160× higher

argmax=signal support is bounded at $mT_{ll} \in [52.8, 202]$ and $m_{ll} \leq 100$ — despite `mtll` **not being an input feature**.

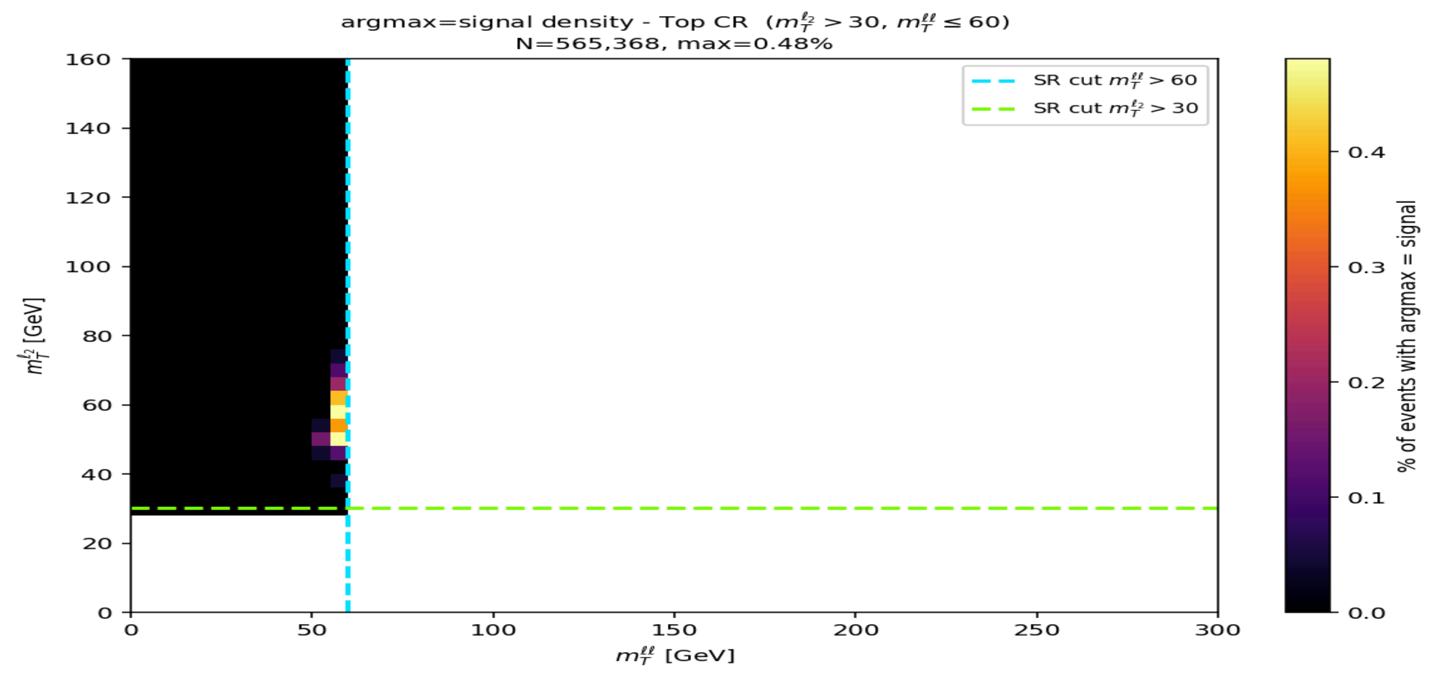
03 · Where the network puts tt̄ — and where the hand-made CR sits

argmax=tt̄ density across the same plane, with the cut-defined top CR overlaid



The tt̄ class fills a **broad** region — the cut box (mT_{ll}≤60 & mT_{l2}>30) sits inside it but captures only **15.3%** of all argmax=tt̄ events. Purity inside the box is 42.5% vs 38.1% outside: the cut buys **+4 points of tt̄ fraction while discarding 85% of the tt̄ the network already identifies.** That is the whole case for the MVA-defined CR on one plot.

03b · Same behaviour in the top CR — separation is learned, not cut



Across the entire cut-defined top CR the argmax=signal fraction peaks at **0.48%**; only **66 of 565,368** events are called signal (0.0117%), max P(H+c) = 0.358 — below the 0.5 needed to ever win argmax.

The cut is redundant with the classifier. The network independently recovers the same region because both follow the same physics — not because it was trained on the cut. `hww_MVA.yaml` has no mT/m_{ll} cut in its base selection; labels are process truth.

04 · Cutting shrinks the training set the MVA needs

Efficiency of each kinematic cut, on top of the ≥1 c-jet requirement

CUT	ε_S	ε_B	$S/\sqrt{B}$	GAIN
$mT_{ll} > 60$	0.926	0.790	0.00091	1.04×
$mT_{l2} > 30$	0.890	0.827	0.00086	0.98×
$m_{ll} \leq 72$	0.978	0.397	0.00136	1.55×
all three (the SR)	**0.845**	**0.271**	**0.00143**	**1.63×**

The cost, made concrete. Putting `mll≤72` into the base selection empties the high-m_{ll} CR *by construction* — a region holding **2.38M events at 83.0% $t\bar{t}$ purity**.
You trade a well-populated constraint region for a 1.63× pre-selection number the MVA does not need.

05 · The real lever: the charm tag, not the kinematics

C-JET EFF — H+C SIGNAL
23.1%
medium WP discards 77% of signal

C-JET EFF — GGH
15.9%
the shape-degenerate competitor

ENRICHMENT H+C / GGH
1.46×
ggH:H+c 681:1 → 467:1

Loose vs medium, measured offline on the untagged tree (5,619 H+c events, `hww_2dcat_nocjet`)

C-TAG OPTION	SIGNAL N	EFF OF ≥1-JET BASE	VS MEDIUM
medium (CvL>0.160, CvB>0.304)	3,244	57.7%	1.00×
loose (CvL>0.054, CvB>0.182)	**5,337**	**95.0%**	**1.65×**
no tag at all	5,619	100.0%	1.73×

Loosening the WP recovers 1.65× the signal — most of what dropping the tag entirely would give (1.73×), while still rejecting light-flavour. Compare the kinematic cuts, worth 1.63× on S/√B but paid for with the control regions.

05b · The missing half: what the looser WP admits — and the answer is *no*

`hww_ctag_compare`, 2022preEE, full MC — four categories over one untagged jet collection, weighted yields

PROCESS	BASE (NO TAG)	MEDIUM WP	LOOSE WP	KIN, NO TAG
H+c (signal)	1.79×	**1.00×**	**1.70×**	1.56×
ggH (degenerate)	2.92×	**1.00×**	**2.67×**	2.16×
tt	1.87×	1.00×	1.55×	0.56×
V+jets	2.39×	1.00×	2.23×	0.34×

SIGNAL GAIN, MEDIUM→LOOSE

1.70×

what slide 05 measured

GGH GAIN, MEDIUM→LOOSE

2.67×

the half that was missing

H+C/GGH ENRICHMENT RETAINED

0.64×

loosening destroys 36% of it

ggH grows 1.57× faster than signal. The charm tag exists to buy H+c-over-ggH enrichment, and loosening the WP gives back **36%** of it. Dropping the tag entirely is worse (0.62×). The acceptance gain on slide 05 is real but it is **not free** — this is the measurement that decides it, and it says **keep the medium WP.**

06 · What this argument does not yet prove

- **~~Loosening the WP admits more ggH too.~~ Now measured — see slide 05b.** ggH rises 2.67× against signal's 1.70×, so the enrichment falls to 0.64×. CvL carries the H+c-vs-ggH separation (AUC 0.731), CvB does not (0.551 ≈ coin flip); a looser WP moves down the CvL axis and admits the degenerate background faster than the signal.
- **Purity is measured here; sensitivity is not.** These are event counts, not a limit. The limit also depends on how systematics act on a larger background.
- **Selection and SF boundaries do not coincide.** The 2D SF scheme bins on `x=CvL/(CvL+CvB(1−CvL))` and `y=1−CvB`; a rectangular WP cut is a different shape in that plane. Already true in production, but worth stating.

> **Slides 01–05 are 2022postEE** (`hww_combine_fixed` MVA tree, raw counts); **slide 05b is 2022preEE** (`hww_ctag_compare` , weighted yields, full MC). The two agree on the signal ratio to within a few percent across different eras, weighting, and code paths — the raw-postEE loose/medium of 1.65× against the weighted-preEE 1.70×.
>
> The **H+c and ggH samples are 100% complete** (7/7 and 4/4 partitions), so the signal, ggH, and enrichment figures on slide 05b are final — they were unchanged between the 174/191 and 186/191 snapshots. The tt and V+jets rows are drawn from 186/191 jobs.

Summary

1. **MVA-defined CRs are purer than cut-defined ones** — 87.9% vs 78.1% on $t\bar{t}$, and 2.8× the events.
2. **The MVA already applies the SR kinematic cuts internally** — 0.0098% vs 11.36% argmax=signal across the mT_{ll} wall, with `mtll` not even an input.
3. **Cutting deletes the sidebands the network learns from**, and empties the CRs you need to fit.
4. **Neither acceptance lever is free.** Medium → loose recovers **1.70×** the signal — comparable to the 1.63× the kinematic cuts buy — but admits ggH at **2.67×**, giving back 36% of the H+c/ggH enrichment. **Keep the medium WP.** The kinematic cuts buy 1.63× on $S/\sqrt{B}$ but empty the CRs. The MVA-defined regions remain the one change that costs nothing.

<footer>Slides 01–05: 2022postEE, v11 6-class MVA [hplusc, higgsbkg, tt, st, diboson, vjets], pooled MC 4,046,127 events, raw · Slide 05b: 2022preEE hww_ctag_compare, weighted yields, full MC · purity = true-process fraction</footer>